Game Proposal for Social Good
After Us

A survival and resource-management game about what we throw away, and what it could become

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Genre: Survival / Resource Management / Puzzle / Educational Social Impact Game
Platform: Web (browser), built with Phaser and JavaScript

1. Game Concept

After Us is a 2D single-player survival game in which the player must stay alive through five levels, each with different environmental conditions, by making smart use of the waste and overlooked natural materials left in a damaged world.

The player starts with almost nothing. Around them lie plastic bottles, packaging, broken electronics, old clothes, and also less familiar natural resources that people rarely think of as useful. Every item can be handled in several ways: keep it in the inventory, reuse it, repair it, recycle it into base materials, or transform it into something new, such as a filter, a building material, a bioplastic, or even something edible. The player can also throw it away, which is fast but has consequences.

While doing this, the player must keep up with world conditions that keep changing: hunger, thirst, air pollution, and the local climate. On each level the player collects resources and builds toward a goal, such as surviving for a set time or reaching a target. Resources are limited, so the central idea of the game is:

> **What looks like waste may still be a resource, but every choice has a cost and a consequence.**

Main goal of the game: to teach players unique, real ways of recycling and reusing trash and overlooked natural resources that most people do not know about. In many survival and crafting games the player follows a familiar chain (build a fire, make a hammer, make an axe) that teaches nothing about the real world. In After Us, the crafting and recycling recipes themselves are the lesson: the player learns methods of turning waste into something useful that they can remember and apply outside the game.

2. Story and Theme

The game takes place several decades after a collapse caused by overconsumption, pollution, and the exhaustion of natural resources. Products were made cheaply and replaced instead of repaired, and the waste piled up faster than the planet could absorb it. The player controls one of the few survivors, living in a small temporary shelter.

The story is told through the environment instead of long cutscenes. Old advertisements, warehouses full of identical products, broken phones, empty shops,

photographs, and handwritten notes show the player how the old world worked and why it failed. Across the levels the survivor moves from a rubbish-covered wasteland toward a place that can sustain a small community, which shows that recovery is possible, but only through many small, deliberate choices.

3. Social Good Theme

3.1 The issue

After Us addresses three connected problems: climate change, pollution (of air, water, and land), and the exhaustion of natural resources caused by unsustainable consumption. It also highlights a hopeful side of the same problem: many materials treated as waste can be reused, repaired, recycled, or turned into green inventions that lower the demand for new raw materials.

3.2 Related UN Sustainable Development Goals

- **SDG 12 – Responsible Consumption and Production.** The main SDG. Players decide whether an object is reused, repaired, recycled, or discarded, and they experience what happens when resources become waste after a single use.
- **SDG 13 – Climate Action.** World conditions such as temperature extremes and polluted air respond to the player's energy choices, for example burning waste for fast energy versus using cleaner alternatives.
- **SDG 3 – Good Health and Well-being.** Bad air quality is a direct survival threat in the game, as it is a health risk in real life.
- **SDG 14 and 15 – Life Below Water and Life on Land.** Discarded plastic ends up in polluted water and soil, which affects water and food availability in the game.
- **SDG 11 – Sustainable Cities and Communities.** The later levels focus on rebuilding a settlement with working water, energy, and recycling systems.

3.3 Why we chose this issue

Waste is something everyone produces every day, yet climate change and pollution often feel too large and distant to be personal. We wanted a game that makes the link between an individual decision and its long-term result tangible. We were also inspired by real green inventions, such as edible seaweed-based packaging and bottles used for solar water disinfection, that show how "trash" can be valuable. These are rarely known by the public.

3.4 Target audience

Our main audience is teenagers and young adults (about 15-25), the generation that will live longest with the consequences of climate change. Casual gamers are a secondary audience: the game is a fun survival game first, so they learn without being lectured.

4. Social Good Integration into Gameplay

The environmental theme is not only the setting. It is the main system the player interacts with. Almost every important decision involves an item and a choice about what to do with it, and each choice changes both the player's survival chances and the state of the world.

- **Keep:** What the player does: Store the item for later (inventory is limited). Trade-off: No cost now, but takes up space and may never be used.
- **Reuse:** What the player does: Use as is, e.g. a bottle becomes a water container. Trade-off: Cheap and quick, but the item is then locked into that role.
- **Repair:** What the player does: Fix a broken object, e.g. a pump or flashlight. Trade-off: Needs materials, energy, and time, but gives a lasting benefit.
- **Recycle:** What the player does: Dismantle and sort into plastic, metal, glass, fabric, electronics, or organics. Trade-off: Slowest option; the yield depends on the sorting puzzle.
- **Transform:** What the player does: Craft a new invention from waste (see 6.5). Trade-off: Requires knowledge, materials, and a workbench; highest long-term payoff.
- **Discard / Extract:** What the player does: Throw away, burn, or take new raw material from nature. Trade-off: Fast and easy now, but raises pollution or depletes resources later.

The most important part of this system is what the player learns from it. The options above are not just generic labels: the game teaches specific, lesser-known ways to reuse and transform waste, so that the main goal of the game (teaching unique recycling methods) is the reward the player is playing for. This system gives us several ways to make the message experiential:

- **Immediate consequences.** When the player burns or dumps something, the effect shows immediately: for example, the sky gets darker and the air worsens. Players will also sometimes find that they need exactly the material they threw away earlier.
- **Real trade-offs.** Recycling takes more time and energy than discarding, so it is never an automatic "correct" answer. The player has to judge what the situation needs now versus later.
- **Discovery as reward.** Learning a new, unusual recipe (for example, purifying water with sunlight and a plastic bottle) gives the player a real reason to look at rubbish differently. This is the core of the game, not an extra.
- **Visible feedback.** The environment changes right away (cleaner water, greener ground, clearer sky, or the opposite, such as a darkening sky) instead of showing only a numerical score.

5. Intended Impact and Message

Main message: Discarded objects do not simply disappear. Their resources and consequences stay in the system, and with creativity, many of them can become something useful again.

By the end of the game, players should have experienced and understood that:

- resources are limited and human activity (consumption, burning, dumping) directly affects air, water, and climate;
- many people do not know about the unusual ways trash can be recycled or reused, and the game's main goal is to teach these methods (unlike typical games where crafting chains such as fire, then hammer, teach nothing real);
- reusing and repairing can be better than recycling, and recycling is better than throwing away, but there are no perfect choices;
- surprising green inventions exist, such as bioplastics, edible packaging, and mushroom-based materials, and everyday waste can be an input for them;
- short-term convenience often causes long-term problems.

By the end, the player should be able to remember and explain at least one or two unusual recycling methods they learned in the game. The shift we want to see in the player's thinking is from "I need resources to survive" to "maybe I shouldn't throw this away" and finally "if I use this well now, I won't need to consume something new later."

How we will know whether the message works

We will test the game with a small group of players (4–10) and ask them to fill out a short survey after playing. The survey will check:

- whether the player can name at least one unusual way waste can be recycled or turned into something useful, and explain how it works;
- whether the player noticed the link between their choices and the changes in the world;
- agreement ratings (1–5) with statements such as "I learned a way to reuse waste that I did not know before" and "The game made me think about what happens to an object after I throw it away."

6. Gameplay and Rules

6.1 Main player actions

Move around the map, explore, collect and inspect objects, manage the inventory, reuse, repair, recycle, and craft items, and respond to the world conditions.

6.2 Resources and world conditions

The player manages two groups of values. Survival needs (such as hunger, thirst, and health) must be kept above zero. World conditions (such as air pollution, temperature, and the overall state of the environment) change over time and depend partly on the

player's choices. The exact set of conditions and how they combine may differ from level to level.

Materials found in the world are the raw inputs. They can be human-made (plastic, glass, metal, fabric, electronics) or natural and less commonly used (for example plants, algae, organic leftovers, or clay). Energy and inventory space are limited, and actions cost energy or time. The final list of materials will be decided during design.

6.3 Progress within a level

On each level the player accumulates resources and builds toward the level goal. The player explores, collects objects, decides what to do with each one, processes materials, and uses the results to survive and to build things that move them closer to the goal. Consequences are visible right away: when the player burns something, the sky becomes darker; when they clean or restore something, the area becomes visibly healthier. The player therefore always sees how their decisions affect the world while they work toward the goal.

6.4 Levels

The game is planned as a series of five levels. Each level places the player in a different setting with its own combination of world conditions, and gives a goal such as surviving for a set time or reaching a target (for example, securing a stable water source). The specific settings, conditions, and goals are still being designed. The general plan is:

- The first level is a gentle introduction to the core loop of finding items and choosing what to do with them, using simple, familiar resources such as plastic waste.
- The middle levels introduce new world conditions and challenges (such as pollution, climate extremes, or scarcity of a key resource), new materials, and new ways of transforming them.
- The final level combines the systems from earlier levels and asks the player to build something lasting, where waste becomes input again.

Between levels, the difficulty grows through fewer easy resources, more complex objects, stronger consequences, and the need to plan further ahead.

6.5 Unique recycling methods and green inventions (main goal)

This is the main goal of the game. Instead of the familiar crafting chains of most games, the player learns real, lesser-known ways of recycling and reusing trash and natural resources. Besides simple reuse, repair, and recycling, the player can discover ways to transform waste and less commonly used natural resources into new things. These are inspired by real-world green ideas, for example bioplastics (including edible ones), simple water purification, natural insulation, composting, or filtering materials. Recipes will be simplified for gameplay, and the final list, the level at which each is unlocked, and how the player learns them (found notes, experimentation, or upgrades) will be decided during development.

6.6 Recycling puzzle

When the player recycles, a short sorting mini-game may appear in which mixed materials must be placed into the correct categories. How well the player sorts would affect how much usable material is recovered, so recycling is a skill and not just a button. The exact rules and numbers are still to be designed.

6.7 Important rules and restrictions

- Inventory space and energy are limited.
- Some materials can only be obtained by recycling, and some objects cannot be repaired.
- Actions take different amounts of time (discard is fastest, recycling is slowest), so the best option depends on the situation.
- Pollution builds up and can make water sources unsafe and reduce food and resource availability, and the change is visible in the world right away.
- Extracting new raw materials from nature is quick but permanently reduces what is available in that area.

6.8 Single-player or multiplayer

Single-player. This keeps the focus on personal decision-making and keeps the scope realistic for a team of two.

6.9 Typical session

A level takes around 15–30 minutes. The player explores the area, collects items, decides what to do with each one, and uses recycling methods and recipes to gain materials and build things. While doing so, they keep survival needs and world conditions under control and work toward the level goal. Along the way the player discovers new, unusual recycling methods and sees the world change in response to their choices.

6.10 Open design decisions

The level settings, world condition details, material list, recipes, and numbers (resource amounts, action times, pollution rates) are not fixed yet and will be developed and balanced during prototyping. We plan to build and test the core loop first, and add world conditions and further levels step by step.

7. Winning, Losing, and Game Objectives

Objective per level: reach the level goal while keeping survival needs under control (see 6.4).

Winning the game: finish the final level by establishing a sustainable settlement with reliable clean water, renewable or low-impact energy, food production, a working recycling loop, and by using the recycling methods they learned along the way. The ending shows a visible but modest recovery of the world, which is improved, not magically fixed.

Losing a level: the player runs out of water or food, health reaches zero (for example from extreme temperatures or polluted air), or the environment becomes too degraded to supply essential resources. Failure comes from the consequences of resource decisions, not from a “bad person” score.

8. Platform and Development Tools

- **Platform:** Web (browser) on desktop. The game runs from a link, so it is easy to share with playtesters and needs no installation.
- **Engine and language:** Phaser (2D game framework) with JavaScript (TypeScript optional) for movement, inventory, resource and world-condition systems, crafting, and progression. We chose it because both team members already know JavaScript.
- **Data and maps:** Items, recipes, and level settings are stored in JSON files separate from the code, so they can be changed easily as the design develops. Maps are built with Tiled.
- **Art:** Aseprite (pixel art) and Krita (2D art), plus free asset packs while prototyping.
- **Audio:** Free or self-made sound effects and music.
- **Tools:** Git and GitHub for version control; GitHub Pages or Netlify to host test builds.

Prototype scope for two students: one playable character, a small number of short levels that can reuse a limited set of maps, a limited set of material types and interactive objects, a basic inventory and crafting system, a recycling mini-game, and a simple environment state system. Extra features are added only once the core is stable.

9. Originality and Existing Games

Survival and crafting games such as Minecraft and Don’t Starve show how engaging resource gathering can be, and This War of Mine and Frostpunk show how hard survival choices create tension. Eco and Terra Nil address ecological topics through systems: Eco models a shared ecosystem, and Terra Nil is about restoring damaged land. There are also many educational recycling games, but they are often simple sorting games with little link to survival.

After Us is different in three ways:

- **Waste as a decision.** Waste is at the center of the survival loop. The player does not only gather from nature; every used object is a decision between waste and resource.
- **Real inventions as gameplay.** Our main goal is to teach unique, real recycling and reuse methods that many people do not know about. Instead of the usual

"build a fire, make a hammer" chains, each recipe is a real green idea, so players learn concrete methods and not only that "pollution is bad."

- **Experience over explanation.** Consequences are visible in the world right away, so the message is learned through experience and not through text.

We do not adapt an existing game. Our contribution is a new combination: a survival-management loop built around reuse, repair, recycling, and transformation, with levels shaped by real climate and pollution problems.